

Medical Omnichannel Intelligence

Databricks FE BAR · Medical Affairs (oncology) · synthetic, non-promotional demo

The business problem

Daiichi Sankyo's oncology franchise (Enhertu, Datroway) is won through scientific exchange between MSLs and oncology KOLs.

Today that exchange is **flying blind**:

- MSL field notes, medical inquiries, congress activity and digital engagement live in **separate systems**.
- MSLs can't see **which KOLs to prioritize, what topics** they care about, or **what questions are unanswered**.
- Result: **missed follow-ups, slow medical-inquiry responses, and high-influence KOLs left under-served** during critical launch windows.

The outcome we deliver

A governed platform that unifies every engagement signal and turns it into a **prioritized, AI-assisted next-best-engagement workflow** in an MSL cockpit.

Estimated ~\$3.4M / year for a 120-MSL field-medical org:

Lever	Estimated annual value
MSL prep time saved (AI pre-engagement briefings, ~2 h/MSL/week)	~\$1.3M
Faster medical-inquiry resolution (overdue surfacing + SLA analytics)	~\$0.6M
Priority-KOL reach uplift (90-day reach ~44% → 60%)	~\$1.5M

- **Executive sponsor (VP Medical Affairs):** higher scientific share-of-voice with priority KOLs; defensible inquiry-SLA governance.
- **Domain owner (Field Medical Director):** every MSL walks in prepared, compliant, and pointed at the highest-need KOLs first.

What the MSL sees

Medical Engagement Copilot (deployed Databricks App):

1. **MSL Action Queue** — KOLs ranked by a priority score (influence × educational need × engagement gap), each with a trigger (“open medical inquiry overdue”), a topic, and a recommended date. Served in milliseconds from **Lakebase**.
2. **Pre-Engagement Briefing** — a click generates a grounded, non-promotional AI briefing for the selected KOL, plus their 360 (interests, inquiries, congress, publications) and a **medically-approved** content recommendation (with review ID).
3. **Write-back** — the MSL logs a review decision; it’s persisted transactionally to Lakebase. Human-in-the-loop, always.

The integrated journey (one governed flow)

Raw synthetic CSVs (UC Volume)

- Lakeflow (Auto Loader) → BRONZE (9 streaming tables)
- SILVER clean + ai_query() extracts structure from free-text MSL notes ← Gen AI
- GOLD HCP 360 · priority · next-best-engagement · SLA · territory (metric views)
 - Genie (natural-language Q&A)
 - Lakebase (operational MSL action queue + review write-back)
 - App (Medical Engagement Copilot)

Unity Catalog governs every object end-to-end (tags, PII class, lineage, scoped grants).

Lakeflow · Unity Catalog · Lakebase · Gen AI · Genie · Databricks App — **integrated, not six disconnected demos.**

Proof it ran (evidence in the repo, as text)

- **Lakeflow:** serverless pipeline `moa_medallion_febar` COMPLETED in 215s; 3,081 synthetic rows → bronze → silver → 13 gold tables.
- **Gen AI:** 500 free-text MSL notes → 500 structured rows; real before/after samples committed; live grounded briefing for the top KOL.
- **Lakebase:** 300-row action queue synced to Postgres; write-back table receiving MSL decisions; least-privilege SP grants.
- **Genie:** live NL question → correct SQL → answer (“SO-03 lowest reach at 26.3%”).
- **App:** deployed, `ComputeState.ACTIVE`, deployment SUCCEEDED.
- **Governance:** UC tag policies enforced; `np_i` classified PII.

(See evidence/01–09.)

Decisions & trade-offs

- **SQL Lakeflow, serverless** — set-based transforms are most legible in SQL; `ai_query` is native. Materialized views give reproducible full refresh over non-deterministic AI.
- **Lakebase for OLTP serving** — Delta scans ≠ interactive cockpit; Lakebase adds millisecond lookups + transactional write-back while UC keeps one governance plane. SNAPSHOT sync fits daily MSL planning; CONTINUOUS is a one-line change.
- **Metric views** — one governed definition of “reach”/“engagement” shared by Genie and the app, so every surface agrees.
- **Human-in-the-loop, non-promotional** — every AI output is a reviewable recommendation, grounded only in the KOL’s own governed signals; approved content only.

AI as a force multiplier

Built end-to-end with Claude (Isaac) driving Databricks MCP tools + CLI: synthetic data, pipeline authoring/run, Lakebase reverse-ETL + write-back, Genie, and app deployment — including live debugging (metric-view YAML, `psycopg`→`pg8000` under Spark serverless, SDK pinning).

Next steps: CONTINUOUS Lakebase sync for real-time queues · AI/BI dashboard on the same metric views · MLflow-evaluated briefing quality · expand beyond oncology to full portfolio.

All data synthetic. Medical Affairs (non-promotional). Recommendations require MSL review.